

25CS1302E - DATABASE SYSTEMS ENGINEERING AND DISTRIBUTED BACKEND DEVELOPMENT

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SECTION: 2B

LAB COURSEWORK

Q. Week 3:

Employee Database Management and SQL Query Operations

A company wants to maintain its employee information using a relational database system. The database should store information about departments, employees, and salary grades and allow the organization to retrieve meaningful information by establishing relationships between these tables.

As a database developer, your task is to create a database named **employee** and design the following tables:

- Dept — containing department number, department name, and location.
- Emp — containing employee number, employee name, job, manager, hire date, salary, commission, and department number.
- Salgrade — containing salary grade, minimum salary, and maximum salary.

Insert the required department, employee, and salary-grade records into the respective tables. Establish the relationship between the Emp and Dept tables using a foreign key.

After creating and populating the tables, perform SQL queries using JOINS, self-JOINS, nested queries, aggregate functions, and subqueries to retrieve information such as employee names with their departments, employees with their salary grades, employee-manager relationships, departments with or without employees, employees earning above specified salary levels, and employees belonging to departments based on specific conditions.

This practical helps develop skills in relational database design, table relationships, JOIN operations, self-JOINs, aggregate functions, correlated subqueries, nested subqueries, and advanced SQL query processing.

```
MYSQL: Last login: Mon Aug 17 10:50:32 on ttys000
poojithamusunuri@Poojithas-MacBook-Air ~ % mysql -u root -p
Enter password:
Welcome to the MySQL monitor.  Commands end with ; or \g.
Your MySQL connection id is 9
Server version: 9.7.1 Homebrew
Copyright (c) 2000, 2026, Oracle and/or its affiliates.
Oracle is a registered trademark of Oracle Corporation and/or its
affiliates. Other names may be trademarks of their respective
owners.
Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.
mysql> CREATE DATABASE employee;
Query OK, 1 row affected (0.032 sec)
mysql> USE employee;
Database changed
mysql> SELECT DATABASE();
+-----+
| DATABASE() |
+-----+
| employee   |
+-----+
1 row in set (0.000 sec)
mysql> CREATE TABLE Dept (
->   deptno INT PRIMARY KEY,
->   dname VARCHAR(20),
->   loc VARCHAR(20)
-> );
Query OK, 0 rows affected (0.014 sec)
mysql> SHOW TABLES;
+-----+
| Tables_in_employee |
+-----+
| Dept                |
+-----+
1 row in set (0.001 sec)
mysql> INSERT INTO Dept VALUES
-> (10, 'Accounting', 'New York'),
-> (20, 'Research', 'Dallas'),
-> (30, 'Sales', 'Chicago'),
```

```
-> (40, 'Operations', 'Boston');
Query OK, 4 rows affected (0.008 sec)
Records: 4 Duplicates: 0 Warnings: 0
mysql> SELECT * FROM Dept;
```

```
+-----+-----+-----+
| deptno | dname   | loc   |
+-----+-----+-----+
| 10 | Accounting | New York |
| 20 | Research   | Dallas   |
| 30 | Sales      | Chicago  |
| 40 | Operations | Boston   |
+-----+-----+-----+
```

4 rows in set (0.000 sec)

```
mysql> CREATE TABLE Emp (
-> empno INT PRIMARY KEY,
-> ename VARCHAR(20),
-> job VARCHAR(20),
-> mgr INT,
-> hiredate DATE,
-> sal DECIMAL(7,2),
-> comm DECIMAL(7,2),
-> deptno INT,
-> FOREIGN KEY (deptno) REFERENCES Dept(deptno)
-> );
```

Query OK, 0 rows affected (0.011 sec)

```
mysql> DESC Emp;
```

```
+-----+-----+-----+-----+-----+-----+
| Field | Type          | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| empno | int           | NO   | PRI | NULL    |      |
| ename | varchar(20)   | YES  |     | NULL    |      |
| job   | varchar(20)   | YES  |     | NULL    |      |
| mgr   | int           | YES  |     | NULL    |      |
| hiredate | date         | YES  |     | NULL    |      |
| sal   | decimal(7,2) | YES  |     | NULL    |      |
| comm  | decimal(7,2) | YES  |     | NULL    |      |
| deptno | int           | YES  | MUL | NULL    |      |
+-----+-----+-----+-----+-----+-----+
```

8 rows in set (0.007 sec)

```
mysql> INSERT INTO Emp
```

```
-> (empno, ename, job, mgr, hiredate, sal, comm, deptno)
-> VALUES
-> (7369, 'SMITH', 'CLERK', 7902, '1980-12-23', 800, NULL, 20),
-> (7499, 'ALLEN', 'SALESMAN', 7698, '1981-02-20', 1600, 300, 30),
```

```

-> (7521, 'WARD', 'SALESMAN', 7698, '1981-02-22', 1250, 500, 30),
-> (7566, 'JONES', 'MANAGER', 7839, '1981-04-02', 2975, NULL, 20),
-> (7654, 'MARTIN', 'SALESMAN', 7698, '1981-09-28', 1250, 1400, 30),
-> (7698, 'BLAKE', 'MANAGER', 7839, '1981-05-01', 2850, NULL, 30),
-> (7782, 'CLARK', 'MANAGER', 7839, '1981-06-09', 2450, NULL, 10),
-> (7788, 'SCOTT', 'ANALYST', 7566, '1982-12-09', 3000, NULL, 20),
-> (7839, 'KING', 'PRESIDENT', NULL, '1981-11-17', 5000, NULL, 10),
-> (7844, 'TURNER', 'SALESMAN', 7698, '1981-09-08', 1500, 0, 30),
-> (7876, 'ADAMS', 'CLERK', 7788, '1983-01-12', 1100, NULL, 20),
-> (7900, 'JAMES', 'CLERK', 7698, '1981-12-03', 950, NULL, 30),
-> (7902, 'FORD', 'ANALYST', 7566, '1981-12-03', 3000, NULL, 20),
-> (7934, 'MILLER', 'CLERK', 7782, '1982-01-23', 1300, NULL, 10);

```

Query OK, 14 rows affected (0.004 sec)

Records: 14 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Emp;

```

+-----+-----+-----+-----+-----+-----+-----+-----+
| empno | ename  | job    | mgr | hiredate | sal   | comm  | deptno |
+-----+-----+-----+-----+-----+-----+-----+-----+
| 7369 | SMITH  | CLERK  | 7902 | 1980-12-23 | 800.00 | NULL | 20 |
| 7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 | 30 |
| 7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00 | 30 |
| 7566 | JONES  | MANAGER | 7839 | 1981-04-02 | 2975.00 | NULL | 20 |
| 7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00 | 30 |
| 7698 | BLAKE  | MANAGER | 7839 | 1981-05-01 | 2850.00 | NULL | 30 |
| 7782 | CLARK  | MANAGER | 7839 | 1981-06-09 | 2450.00 | NULL | 10 |
| 7788 | SCOTT  | ANALYST | 7566 | 1982-12-09 | 3000.00 | NULL | 20 |
| 7839 | KING   | PRESIDENT | NULL | 1981-11-17 | 5000.00 | NULL | 10 |
| 7844 | TURNER | SALESMAN | 7698 | 1981-09-08 | 1500.00 | 0.00 | 30 |
| 7876 | ADAMS  | CLERK   | 7788 | 1983-01-12 | 1100.00 | NULL | 20 |
| 7900 | JAMES  | CLERK   | 7698 | 1981-12-03 | 950.00 | NULL | 30 |
| 7902 | FORD   | ANALYST | 7566 | 1981-12-03 | 3000.00 | NULL | 20 |
| 7934 | MILLER | CLERK   | 7782 | 1982-01-23 | 1300.00 | NULL | 10 |
+-----+-----+-----+-----+-----+-----+-----+-----+

```

14 rows in set (0.002 sec)

mysql> CREATE TABLE Salgrade (

```

->   grade INT,
->   losal INT,
->   hisal INT
-> );

```

Query OK, 0 rows affected (0.007 sec)

mysql> INSERT INTO Salgrade VALUES

```

-> (1, 700, 1200),
-> (2, 1201, 1400),
-> (3, 1401, 2000),

```

-> (4, 2001, 3000),

-> (5, 3001, 9999);

Query OK, 5 rows affected (0.003 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM Salgrade;

```
+-----+-----+-----+
| grade | losal | hisal |
+-----+-----+-----+
| 1 | 700 | 1200 |
| 2 | 1201 | 1400 |
| 3 | 1401 | 2000 |
| 4 | 2001 | 3000 |
| 5 | 3001 | 9999 |
+-----+-----+-----+
```

5 rows in set (0.000 sec)

mysql> SHOW TABLES;

```
+-----+
| Tables_in_employee |
+-----+
| Dept                |
| Emp                 |
| Salgrade            |
+-----+
```

3 rows in set (0.002 sec)

mysql> SELECT * FROM Dept;

```
+-----+-----+-----+
| deptno | dname      | loc   |
+-----+-----+-----+
| 10 | Accounting | New York |
| 20 | Research   | Dallas  |
| 30 | Sales      | Chicago |
| 40 | Operations | Boston  |
+-----+-----+-----+
```

4 rows in set (0.000 sec)

mysql> SELECT * FROM Emp;

```
+-----+-----+-----+-----+-----+-----+-----+
| empno | ename  | job      | mgr | hiredate | sal   | comm  | deptno |
+-----+-----+-----+-----+-----+-----+-----+
| 7369 | SMITH  | CLERK    | 7902 | 1980-12-23 | 800.00 | NULL | 20 |
| 7499 | ALLEN  | SALESMAN | 7698 | 1981-02-20 | 1600.00 | 300.00 | 30 |
| 7521 | WARD   | SALESMAN | 7698 | 1981-02-22 | 1250.00 | 500.00 | 30 |
| 7566 | JONES  | MANAGER  | 7839 | 1981-04-02 | 2975.00 | NULL | 20 |
| 7654 | MARTIN | SALESMAN | 7698 | 1981-09-28 | 1250.00 | 1400.00 | 30 |
| 7698 | BLAKE  | MANAGER  | 7839 | 1981-05-01 | 2850.00 | NULL | 30 |
```

7782	CLARK		MANAGER		7839		1981-06-09		2450.00		NULL		10	
7788	SCOTT		ANALYST		7566		1982-12-09		3000.00		NULL		20	
7839	KING		PRESIDENT		NULL		1981-11-17		5000.00		NULL		10	
7844	TURNER		SALESMAN		7698		1981-09-08		1500.00		0.00		30	
7876	ADAMS		CLERK		7788		1983-01-12		1100.00		NULL		20	
7900	JAMES		CLERK		7698		1981-12-03		950.00		NULL		30	
7902	FORD		ANALYST		7566		1981-12-03		3000.00		NULL		20	
7934	MILLER		CLERK		7782		1982-01-23		1300.00		NULL		10	

+-----+-----+-----+-----+-----+-----+-----+-----+

14 rows in set (0.001 sec)

mysql> SELECT * FROM Salgrade;

+-----+-----+-----+						
	grade		losal		hisal	
+-----+-----+-----+						
	1		700		1200	
	2		1201		1400	
	3		1401		2000	
	4		2001		3000	
	5		3001		9999	
+-----+-----+-----+						

5 rows in set (0.000 sec)

mysql> SELECT e.ename, d.dname
-> FROM Emp e
-> JOIN Dept d
-> ON e.deptno = d.deptno;

+-----+-----+				
	ename		dname	
+-----+-----+				
	CLARK		Accounting	
	KING		Accounting	
	MILLER		Accounting	
	SMITH		Research	
	JONES		Research	
	SCOTT		Research	
	ADAMS		Research	
	FORD		Research	
	ALLEN		Sales	
	WARD		Sales	
	MARTIN		Sales	
	BLAKE		Sales	
	TURNER		Sales	
	JAMES		Sales	
+-----+-----+				

14 rows in set (0.001 sec)

```
mysql> SELECT e.ename, e.sal, s.grade
-> FROM Emp e
-> JOIN Salgrade s
-> ON e.sal BETWEEN s.losal AND s.hisal;
```

```
+-----+-----+-----+
| ename | sal   | grade |
+-----+-----+-----+
| SMITH | 800.00 | 1 |
| ALLEN | 1600.00 | 3 |
| WARD  | 1250.00 | 2 |
| JONES | 2975.00 | 4 |
| MARTIN | 1250.00 | 2 |
| BLAKE | 2850.00 | 4 |
| CLARK | 2450.00 | 4 |
| SCOTT | 3000.00 | 4 |
| KING  | 5000.00 | 5 |
| TURNER | 1500.00 | 3 |
| ADAMS | 1100.00 | 1 |
| JAMES | 950.00 | 1 |
| FORD  | 3000.00 | 4 |
| MILLER | 1300.00 | 2 |
```

```
+-----+-----+-----+
14 rows in set (0.000 sec)
```

```
mysql> SELECT e.ename, d.dname, s.grade
-> FROM Emp e
-> JOIN Dept d
-> ON e.deptno = d.deptno
-> JOIN Salgrade s
-> ON e.sal BETWEEN s.losal AND s.hisal;
```

```
+-----+-----+-----+
| ename | dname   | grade |
+-----+-----+-----+
| SMITH | Research | 1 |
| ALLEN | Sales    | 3 |
| WARD  | Sales    | 2 |
| JONES | Research | 4 |
| MARTIN | Sales    | 2 |
| BLAKE | Sales    | 4 |
| CLARK | Accounting | 4 |
| SCOTT | Research | 4 |
| KING  | Accounting | 5 |
| TURNER | Sales    | 3 |
| ADAMS | Research | 1 |
| JAMES | Sales    | 1 |
```

FORD	Research	4	
MILLER	Accounting	2	

+-----+-----+-----+

14 rows in set (0.001 sec)

```
mysql> SELECT e.ename AS Employee,
->      m.ename AS Manager
-> FROM Emp e
-> LEFT JOIN Emp m
-> ON e.mgr = m.empno;
```

+-----+-----+

Employee	Manager	
----------	---------	--

+-----+-----+

SMITH	FORD	
ALLEN	BLAKE	
WARD	BLAKE	
JONES	KING	
MARTIN	BLAKE	
BLAKE	KING	
CLARK	KING	
SCOTT	JONES	
KING	NULL	
TURNER	BLAKE	
ADAMS	SCOTT	
JAMES	BLAKE	
FORD	JONES	
MILLER	CLARK	

+-----+-----+

14 rows in set (0.001 sec)

```
mysql> SELECT d.dname,
->      e.ename AS Manager
-> FROM Dept d
-> JOIN Emp e
-> ON d.deptno = e.deptno
-> WHERE e.job = 'MANAGER';
```

+-----+-----+

dname	Manager	
-------	---------	--

+-----+-----+

Research	JONES	
Sales	BLAKE	
Accounting	CLARK	

+-----+-----+

3 rows in set (0.002 sec)

```
mysql> SELECT d.dname,
->      e.ename AS Manager,
```



```

->      s.grade
-> FROM Dept d
-> JOIN Emp e
-> ON d.deptno = e.deptno
-> AND e.job = 'MANAGER'
-> JOIN Salgrade s
-> ON e.sal BETWEEN s.losal AND s.hisal
-> ORDER BY s.grade;

```

```

+-----+-----+-----+
| dname   | Manager | grade |
+-----+-----+-----+
| Research | JONES   | 4     |
| Sales    | BLAKE   | 4     |
| Accounting | CLARK   | 4     |
+-----+-----+-----+

```

3 rows in set (0.001 sec)

```

mysql> SELECT e.ename,
->      e.comm,
->      s.grade
-> FROM Emp e
-> JOIN Salgrade s
-> ON e.sal BETWEEN s.losal AND s.hisal
-> WHERE e.comm IS NOT NULL;

```

```

+-----+-----+-----+
| ename   | comm    | grade |
+-----+-----+-----+
| ALLEN   | 300.00  | 3     |
| WARD    | 500.00  | 2     |
| MARTIN  | 1400.00 | 2     |
| TURNER  | 0.00    | 3     |
+-----+-----+-----+

```

4 rows in set (0.001 sec)

```

mysql> SELECT e.ename AS Manager_Employee,
->      m.ename AS Reports_To
-> FROM Emp e
-> JOIN Emp m
-> ON e.mgr = m.empno
-> WHERE e.job = 'MANAGER';

```

```

+-----+-----+-----+
| Manager_Employee | Reports_To |
+-----+-----+-----+
| JONES            | KING      |
| BLAKE            | KING      |
| CLARK            | KING      |
+-----+-----+-----+

```

+-----+-----+

3 rows in set (0.001 sec)

mysql> SELECT e.ename

-> FROM Emp e

-> JOIN Emp m

-> ON e.mgr = m.empno

-> WHERE e.deptno = m.deptno;

+-----+

| ename |

+-----+

| SMITH |

| ALLEN |

| WARD |

| MARTIN |

| CLARK |

| SCOTT |

| TURNER |

| ADAMS |

| JAMES |

| FORD |

| MILLER |

+-----+

11 rows in set (0.001 sec)

mysql> SELECT e.ename

-> FROM Emp e

-> JOIN Emp m

-> ON e.mgr = m.empno

-> WHERE e.deptno <> m.deptno;

+-----+

| ename |

+-----+

| JONES |

| BLAKE |

+-----+

2 rows in set (0.001 sec)

mysql> SELECT e.ename, d.dname

-> FROM Emp e

-> JOIN Dept d

-> ON e.deptno = d.deptno

-> WHERE UPPER(SUBSTRING(e.ename, 1, 1))

-> = UPPER(SUBSTRING(d.dname, 1, 1));

Empty set (0.002 sec)

mysql> SELECT e.ename, e.hiredate, s.grade

-> FROM Emp e

```

-> JOIN Salgrade s
-> ON e.sal BETWEEN s.losal AND s.hisal
-> WHERE MONTH(e.hiredate) = MONTH(CURDATE())
-> AND s.grade = MOD(YEAR(e.hiredate), 10);

```

Empty set (0.003 sec)

```
mysql> SELECT e.ename
```

```

-> FROM Emp e
-> JOIN Salgrade s
-> ON e.sal BETWEEN s.losal AND s.hisal
-> WHERE MOD(e.empno, 2) = MOD(e.mgr, 2)
-> AND MOD(e.empno, 2) = MOD(s.grade, 2);

```

```

+-----+
| ename |
+-----+
| MARTIN |
| SCOTT |
| FORD |
| MILLER |
+-----+

```

4 rows in set (0.001 sec)

```
mysql> SELECT e.ename,
```

```

-> e.deptno,
-> s.grade
-> FROM Emp e
-> JOIN Salgrade s
-> ON e.sal BETWEEN s.losal AND s.hisal
-> WHERE s.grade = FLOOR(e.deptno / 10);

```

```

+-----+-----+-----+
| ename | deptno | grade |
+-----+-----+-----+
| ALLEN | 30 | 3 |
| TURNER | 30 | 3 |
+-----+-----+-----+

```

2 rows in set (0.002 sec)

```
mysql> SELECT e.ename,
```

```

-> d.dname,
-> d.loc
-> FROM Emp e
-> JOIN Dept d
-> ON e.deptno = d.deptno
-> WHERE LENGTH(e.ename) = LENGTH(d.dname)
-> OR LENGTH(d.dname) = LENGTH(d.loc)
-> OR LENGTH(e.ename) = LENGTH(d.loc);

```

```

+-----+-----+-----+

```

ename	dname	loc
ALLEN	Sales	Chicago
BLAKE	Sales	Chicago
JAMES	Sales	Chicago

3 rows in set (0.001 sec)

```
mysql> SELECT e.ename
-> FROM Emp e
-> JOIN Emp m
-> ON e.mgr = m.empno
-> WHERE e.job = 'CLERK'
-> AND m.job = 'ANALYST';
```

ename
SMITH
ADAMS

2 rows in set (0.001 sec)

```
mysql> SELECT e.ename AS Employee,
-> m.ename AS Manager
-> FROM Emp e
-> JOIN Emp m
-> ON e.mgr = m.empno
-> JOIN Salgrade se
-> ON e.sal BETWEEN se.losal AND se.hisal
-> JOIN Salgrade sm
-> ON m.sal BETWEEN sm.losal AND sm.hisal
-> WHERE se.grade = sm.grade;
```

Employee	Manager
SCOTT	JONES
FORD	JONES

2 rows in set (0.002 sec)

```
mysql> SELECT e.ename
-> FROM Emp e
-> JOIN Emp m
-> ON e.mgr = m.empno
-> WHERE e.hiredate < m.hiredate;
```

ename

+-----+

| SMITH |

| ALLEN |

| WARD |

| JONES |

| BLAKE |

| CLARK |

+-----+

6 rows in set (0.001 sec)

mysql> SELECT ename, sal

-> FROM Emp

-> WHERE sal >

-> (

-> SELECT sal

-> FROM Emp

-> WHERE ename = 'ALLEN'

->);

+-----+-----+

| ename | sal |

+-----+-----+

| JONES | 2975.00 |

| BLAKE | 2850.00 |

| CLARK | 2450.00 |

| SCOTT | 3000.00 |

| KING | 5000.00 |

| FORD | 3000.00 |

+-----+-----+

6 rows in set (0.003 sec)

mysql> SELECT ename, deptno

-> FROM Emp

-> WHERE deptno IN

-> (

-> SELECT deptno

-> FROM Emp

-> WHERE ename = 'SMITH'

->);

+-----+-----+

| ename | deptno |

+-----+-----+

| SMITH | 20 |

| JONES | 20 |

| SCOTT | 20 |

| ADAMS | 20 |

| FORD | 20 |

+-----+-----+

5 rows in set (0.001 sec)

mysql> SELECT ename, sal

-> FROM Emp

-> WHERE sal > ANY

-> (

-> SELECT sal

-> FROM Emp

-> WHERE job = 'SALESMAN'

->);

+-----+-----+

| ename | sal |

+-----+-----+

| ALLEN | 1600.00 |

| JONES | 2975.00 |

| BLAKE | 2850.00 |

| CLARK | 2450.00 |

| SCOTT | 3000.00 |

| KING | 5000.00 |

| TURNER | 1500.00 |

| FORD | 3000.00 |

| MILLER | 1300.00 |

+-----+-----+

9 rows in set (0.004 sec)

mysql> SELECT ename, sal

-> FROM Emp

-> WHERE sal > ALL

-> (

-> SELECT sal

-> FROM Emp

-> WHERE job = 'CLERK'

->);

+-----+-----+

| ename | sal |

+-----+-----+

| ALLEN | 1600.00 |

| JONES | 2975.00 |

| BLAKE | 2850.00 |

| CLARK | 2450.00 |

| SCOTT | 3000.00 |

| KING | 5000.00 |

| TURNER | 1500.00 |

| FORD | 3000.00 |

+-----+-----+

8 rows in set (0.001 sec)

```
mysql> SELECT e1.ename, e1.sal, e1.deptno
```

```
-> FROM Emp e1
```

```
-> WHERE e1.sal >
```

```
-> (
```

```
->   SELECT AVG(e2.sal)
```

```
->   FROM Emp e2
```

```
->   WHERE e2.deptno = e1.deptno
```

```
-> );
```

```
+-----+-----+-----+
| ename | sal   | deptno |
+-----+-----+-----+
| ALLEN | 1600.00 | 30 |
| JONES | 2975.00 | 20 |
| BLAKE | 2850.00 | 30 |
| SCOTT | 3000.00 | 20 |
| KING  | 5000.00 | 10 |
| FORD  | 3000.00 | 20 |
+-----+-----+-----+
```

6 rows in set (0.001 sec)

```
mysql> SELECT d.dname
```

```
-> FROM Dept d
```

```
-> WHERE d.deptno IN
```

```
-> (
```

```
->   SELECT e.deptno
```

```
->   FROM Emp e
```

```
-> );
```

```
+-----+
| dname   |
+-----+
| Accounting |
| Research  |
| Sales     |
+-----+
```

3 rows in set (0.001 sec)

```
mysql> SELECT d.dname
```

```
-> FROM Dept d
```

```
-> WHERE d.deptno NOT IN
```

```
-> (
```

```
->   SELECT e.deptno
```

```
->   FROM Emp e
```

```
-> );
```

```
+-----+
| dname   |
+-----+
```

```

+-----+
| Operations |
+-----+
1 row in set (0.001 sec)
mysql> SELECT ename,
->      (
->      SELECT dname
->      FROM Dept d
->      WHERE d.deptno = e.deptno
->      ) AS Dept_name
-> FROM Emp e;

```

```

+-----+-----+
| ename | Dept_name |
+-----+-----+
| SMITH | Research  |
| ALLEN | Sales     |
| WARD  | Sales     |
| JONES | Research  |
| MARTIN | Sales     |
| BLAKE | Sales     |
| CLARK | Accounting|
| SCOTT | Research  |
| KING  | Accounting|
| TURNER | Sales     |
| ADAMS | Research  |
| JAMES | Sales     |
| FORD  | Research  |
| MILLER | Accounting|

```

```

+-----+-----+
14 rows in set (0.001 sec)
mysql> SELECT deptno, avg_sal
-> FROM
-> (
-> SELECT deptno,
->      AVG(sal) AS avg_sal
-> FROM Emp
-> GROUP BY deptno
-> ) AS dept_avg
-> WHERE avg_sal > 2000;

```

```

+-----+-----+
| deptno | avg_sal |
+-----+-----+
| 10 | 2916.666667 |
| 20 | 2175.000000 |

```



```
+-----+-----+
2 rows in set (0.002 sec)
mysql> SELECT ename
-> FROM Emp
-> WHERE deptno =
-> (
->   SELECT deptno
->   FROM Dept
->   WHERE loc = 'New York'
-> );
```

```
+-----+
| ename |
+-----+
| CLARK |
| KING  |
| MILLER|
+-----+
```

```
3 rows in set (0.001 sec)
mysql>
```